PROJECT BLUEPRINT DOCUMENT

Software Requirements Specification

AI Resume Analyzer

(Resume Intelligence Engine)

Standardized Compliance: IEEE Std 830-1998 Structure

Version: 1.0.0

Date: July 2026

Target Audience: Core Engineering, Project Managers, QA Evaluators

1. Introduction

1.1 Purpose

This document specifies the complete, definitive software requirements for the **AI Resume Analyzer** (internally designated as the *Resume Intelligence Engine*). It outlines the definitive functional modules, operational boundaries, data schemas, communication protocols, and non-functional requirements governing the platform's execution layer.

1.2 Document Conventions

- Requirements are written with deterministic clarity using standard behavioral imperatives (**SHALL**, **SHOULD**, **MAY**).
- Architectural metrics, structural layouts, and conventions strictly follow the standard IEEE 830 system definitions.
- Technical acronyms—including but not limited to *JWT*, *ATS*, *CORS*, *REST*, and *JPA*—align fully with standard industry specifications.

1.3 Intended Audience

This specification is intentionally organized for cross-functional distribution among:

- **Backend and Frontend Engineers:** For building functional system modules, interface controls, and internal routing behaviors.
- **DevOps/Infrastructure Teams:** For orchestration, pipeline provisioning, continuous integration, and secure credential scoping.
- **QA Engineers & Testers:** For mapping robust automated testing suites, black-box validation, and boundary exception suites.
- **Product Managers & Evaluators:** For verifying scope completeness and functional implementation matches.

1.4 Product Scope

The **AI Resume Analyzer** is an automated, enterprise-grade web intelligence tool engineered to dismantle the systemic opacity of corporate Applicant Tracking Systems (ATS). The application processes unstructured text data from an ingested PDF resume alongside a target corporate job description. Utilizing specialized prompt sequencing routed to Large Language Models (LLMs) via the OpenRouter framework, the system provides multi-dimensional match processing.

Upon evaluation, the core engine produces a normalized ATS compliance score, identifies explicit keyword alignment gaps, structural anomalies, and text formatting vulnerabilities. It generates a tailored learning roadmap to address skill gaps, outlines direct career adjustments, lists targeted technical interview preparation questions, and dynamically creates an absolute offline-ready verified verification report (PDF format).

2. Overall Description

2.1 Product Perspective

The system is architected as an isolated, fully decoupled client-server platform optimized for high availability, atomic operations, and responsive state handling. The system boundaries comprise four fundamental segments:

- **Presentation Layer (Client):** A responsive Single Page Application (SPA) driven by Vite, React, Tailwind CSS, and Axios, incorporating rich charting engines (Chart.js) and micro-interactions.
- **Application Layer (Server):** A high-throughput Spring Boot REST API utilizing Spring Security for stateless context handling and Spring Data JPA for object-relational abstraction mappings.
- **Data Layer (Persistence):** A relational PostgreSQL database instance orchestrating normalized data tracking for profiles, metadata, and historical records.
- **External Integration Layer:** Remote infrastructure bridging secure connections to the OpenRouter LLM gateway and rendering local compilation packages (OpenPDF).

2.2 Product Functions

The principal system features managed by the core runtime include:

> **Identity Protection:** Implements secure sign-up, sessionless token creation (JWT), and automated filter-chain route blocks.

> **Ingestion & Parsing Pipeline:** Handles text extraction from multi-part PDF byte streams, isolating unstructured text blocks safely.

> **Semantic AI Profiling:** Drives target prompt generation to map multi-layered keyword scoring matrices via natural language models.

> **Time-Series Historical Analytics:** Tracks chronological evolution metrics for user evaluation profiles to render interactive charts.

2.3 User Classes and Characteristics

- **Job Seekers / Candidates (Primary Class):** Demands clean, low-latency UI performance, straightforward navigation controls, data-dense summaries, and instant report distribution tools.
- **System Administrators (Operational Class):** Requires strict data access, complete application error logs, API configuration secret overrides, and connection status analytics panels.

2.4 Operating Environment

The operational environment boundaries are specified as follows:

- **Client Layer:** Executable within any standard modern web browser engine (Chromium, Gecko, WebKit) with full support for ES6+ scripting models.
- **Server Target:** Platform-agnostic runtime execution matching standard Java Runtime Environments (JRE v17 or higher).
- **Database Node:** Relational instance provisioning matching PostgreSQL v12 or higher specifications.

3. External Interface Requirements

3.1 User Interfaces

The client visual application shall strictly conform to the following graphic tokens:

- **Theming & Branding:** Glassmorphic layout windows utilizing subtle background blurring, semi-opaque borders, and premium modern blue-to-purple gradient backdrops.
- **Interactivity Triggers:** Interactive drag-and-drop ingestion components built with contextual hover transformations, asset byte counters, and validation state indicators.
- **Data Representation:** Custom SVG circular indicators displaying real-time responsive scoring metrics, coupled with detailed radar and trend line graphs.

3.2 Software Interfaces

- **Database Connectors:** High-performance connection lifecycle pooling via `HikariCP` frameworks linked through standard JDBC interfaces.
- **Security Layer:** Custom Spring Security implementation introducing an isolated token validation filter extending `OncePerRequestFilter`.

- **AI Gateway Integration:** Secure REST interactions over standard HTTPS protocol, referencing JSON exchange objects bounded by API schema definitions.

3.3 Communications Interfaces

Cross-Origin Resource Sharing (CORS) rules must explicitly permit the following production domains within the security filter chain configuration:

- `http://localhost:5173` (Local Development Context)
- `https://ai-resume-analyzer-chi-silk.vercel.app` (Production Ingress)

Communication payloads shall exclusively enforce `application/json` formats for standard parameter calls and `multipart/form-data` schemas for binary file uploads.

4. System Features

4.1 User Authentication and Session Management

Users must register using a unique email address, a full name, and a secure password. Session monitoring is managed entirely via stateless JSON Web Tokens (JWT), ensuring strong validation boundaries.

Endpoints:

- `POST /api/users/register` (Public access context)
- `POST /api/users/login` (Public access context)

Passwords must be hashed using the standard BCrypt encryption algorithm prior to permanent database commit actions.

4.2 Resume Upload and Analysis Pipeline

Authorized sessions can pass a raw PDF file alongside a text-based job description block. The underlying service handles document parsing, establishes connection calls to OpenRouter, isolates specific metrics fields, and populates the client layout views.

Endpoints:

- `POST /api/resume/upload` (Authorized JWT scope required)

The transmission schema requires formal `multipart/form-data` structure containing the specific binary file field `resume`, text block `jobDescription`, and structural string parameter `userEmail`.

5. Database Schema & Technical Blueprint

5.1 Data Dictionary & Table Schemas

The persistent layer maintains the three essential structural tables detailed below:

1. users Table

Maintains core user profile credentials and lookup indices.

Column Name	Data Type	Constraints	Description
`id`	BIGINT	PRIMARY KEY, AUTO_INCREMENT	System wide unique entity index.
`name`	VARCHAR(255)	NOT NULL	The user's declared full profile name.
`email`	VARCHAR(255)	NOT NULL, UNIQUE	Primary security identifier parameter.
`password`	VARCHAR(255)	NOT NULL	Secure BCrypt cryptographically hashed string.

2. resume_analysis Table

Maintains persistent operational results generated from the OpenRouter AI evaluation framework.

Column Name	Data Type	Constraints	Description
`id`	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the evaluation log.
`user_email`	VARCHAR(255)	NOT NULL	Maps ownership reference bounds back to the user record.
`resume_score`	INT	NOT NULL, CHECK (0-100)	Calculated numerical ATS matching metric percentage.
`matching_skills`	TEXT	NULLABLE	Isolated matching keywords structured string array.
`missing_skills`	TEXT	NULLABLE	Detected structural gap competencies identified by LLM.
`ai_suggestions`	TEXT	NULLABLE	Targeted copy modifications and layout suggestions.
`learning_roadmap`	TEXT	NULLABLE	Chronological upskilling matrix text content blocks.

6. Non-Functional Requirements & Error States

6.1 Security Mandates

- **Session Expiration:** Issued web authorization tokens (JWT) shall maintain a hard expiration time-to-live limit of 24 hours.
- **Transport Enforcements:** Every programmatic interface call in live environments must enforce standard TLS 1.3 parameters over HTTPS.

6.2 System Exception Vector Maps

Exception Event Triggers	HTTP Status Code	Diagnostic Error JSON Payload
Expired / Malformed Token	`401 Unauthorized`	`{"error": "Signature Validation Failure", "code": "AUTH_ERR_04"}`
Asset Size Ingestion Overflow (>5MB)	`413 Payload Too Large`	`{"error": "Target PDF exceeds system capacity constraint.", "code": "FILE_SIZE_ERR"}`
Upstream Remote LLM Timeout	`504 Gateway Timeout`	`{"error": "OpenRouter did not process the analysis payload.", "code": "LLM_TIMEOUT"}`
Email Registration Collision	`409 Conflict`	`{"error": "An account is already linked to this email address.", "code": "REG_CONFLICT"}`

6.3 Performance Indicators

- **Parsing Performance:** The extraction workflow must distill raw multi-part text content within 1500ms of ingestion arrival.
- **Analytics Delivery:** Aggregated historical logs requested by client view triggers must deliver responsive payloads inside 400ms bounds.